



Elastic Collisions in One Dimension

· Mark Scheme

Further Mechanics 1 Chapter 4: Elastic Collisions in One Dimension. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Newton's law of restitution

1 $e = \frac{5.5 - 4}{7 - 3}$ M1; $e = 0.375$ A1.

2 $2(6) = 2v_A + 4v_B$ M1; $v_B - v_A = 0.5(6)$ M1; $v_A = 0$, $v_B = 3 \text{ ms}^{-1}$ A1.

3 Positive in A's direction: $3(4) + 1(-2) = 3v_A + v_B$ M1; $v_B - v_A = 0.5(4 + 2) = 3$ M1; $v_A = 1.75 \text{ ms}^{-1}$, $v_B = 4.75 \text{ ms}^{-1}$, both in A's original direction A1.

4 $mu = mv_A + 2mv_B$ M1; $v_B - v_A = eu$ M1; $v_B = \frac{1}{3}u(1 + e)$ A1; $v_A = \frac{1}{3}u(1 - 2e)$, which is negative when $e > \frac{1}{2}$, so A reverses A1.

5(a) Positive in P's direction: $2m(2u) - mu = 2mv_P + mv_Q$ M1; $v_Q - v_P = e(2u + u) = 3eu$ M1; $3u = 3v_P + 3eu$, so $v_P = u(1 - e)$ A1; $v_Q = u(1 - e) + 3eu = u(1 + 2e)$ A1.

5(b) $u(1 + 2e) = 4u(1 - e)$ M1 A1; $e = \frac{1}{2}$ A1.

5(c) $v_Q = 2u$; $I = m(2u) - m(-u)$ M1; $= 3mu$ A1.

Section B – Fixed surfaces and loss of kinetic energy

6 $v = \sqrt{2(9.8)(2.5)} = 7 \text{ ms}^{-1}$ M1; rebound speed $0.4 \times 7 = 2.8 \text{ ms}^{-1}$ A1.

7 $e = \frac{6}{10} = 0.6$ B1; $\frac{1}{2}(0.2)(10^2) - \frac{1}{2}(0.2)(6^2)$ M1; $= 6.4 \text{ J}$ A1.

8 $3(5) = 5v$, $v = 3 \text{ ms}^{-1}$ M1; $\frac{1}{2}(3)(5^2) - \frac{1}{2}(5)(3^2)$ M1; $= 15 \text{ J}$ A1.

9 $10 = 2v_A + 3v_B$ and $v_B - v_A = 3$ M1; $v_A = 0.2 \text{ ms}^{-1}$, $v_B = 3.2 \text{ ms}^{-1}$ A1; KE loss $= 25 - \left(\frac{1}{2}(2)(0.2^2) + \frac{1}{2}(3)(3.2^2)\right)$ M1; $= 9.6 \text{ J}$ A1.

10(a) $\sqrt{2(9.8)(2.5)} = 7 \text{ ms}^{-1}$ B1.

10(b) Rebound speed $0.7 \times 7 = 4.9 \text{ ms}^{-1}$ M1; $0 = 4.9^2 - 2(9.8)h$ M1; $h = 1.225 = 1.23 \text{ m}$ A1.

10(c) $\frac{1}{2}(0.2)(7^2 - 4.9^2)$ M1; $= 2.50 \text{ J}$ A1.

10(d) $-4.9 = 4.9 - 9.8t$ M1; $t = 1 \text{ s}$ A1.

Section C – Successive collisions

11 $0.5 \times 2 = 1 \text{ ms}^{-1}$ B1; B now moves towards A, and A is still moving towards B, so they must collide B1.

12 $1(4) = v_A + v_B$ M1; $v_B - v_A = 0.5(4) = 2$ M1; $v_A = 1 \text{ ms}^{-1}$, $v_B = 3 \text{ ms}^{-1}$ A1.

13 $1(3) = w_B + 2w_C$ and $w_C - w_B = 0.5(3)$ M1; $w_B = 0$, $w_C = 1.5 \text{ ms}^{-1}$ A1; B is at rest and A is still moving towards it at 1 ms^{-1} , so they collide again B1.

14 $6 = v_A + 2v_B$, $v_B - v_A = 0.8(6)$ M1; $v_A = -1.2$ (so A moves away from the wall at 1.2 ms^{-1}), $v_B = 3.6 \text{ ms}^{-1}$ A1; B rebounds at $0.25 \times 3.6 = 0.9 \text{ ms}^{-1}$ B1; $0.9 < 1.2$, so B cannot catch A: no further collision A1.

15(a) $6 = v_A + 2v_B$, $v_B - v_A = \frac{1}{4}(6)$ M1 M1; $v_A = 1 \text{ ms}^{-1}$, $v_B = 2.5 \text{ ms}^{-1}$ A1.

15(b) $2(2.5) = 2w_B + 3w_C$, $w_C - w_B = \frac{1}{4}(2.5)$ M1 M1; $w_B = 0.625 \text{ ms}^{-1}$, $w_C = 1.25 \text{ ms}^{-1}$ A1.

15(c) All three move in the same direction; A has speed $1 \text{ ms}^{-1} > 0.625 \text{ ms}^{-1}$ M1; so A catches up with B and collides with it again A1.



Roar-level challenge

16 $A \rightarrow B$: $v_A = \frac{1}{3}u(1-2e)$, $v_B = \frac{1}{3}u(1+e)$ $M1$ $A1$; $B \rightarrow C$: $2mv_B = 2mw_B + 3mw_C$, $w_C - w_B = ev_B$, so $w_B = \frac{1}{5}v_B(2-3e) = \frac{1}{15}u(1+e)(2-3e)$ $M1$ $A1$; A catches B iff $v_A > w_B$: $5(1-2e) > (1+e)(2-3e) \iff e^2 - 3e + 1 > 0 \iff e < \frac{1}{2}(3 - \sqrt{5})$, since $e \leq 1 < \frac{1}{2}(3 + \sqrt{5})$ $A1$.